

Romanization Flattens the Field: Tokenizer-Independent and Downstream Evidence of Script Fragility in Sinhala

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Abstract

Sinhala is written two ways online. Formal text uses the Sinhala script, while everyday messaging is dominated by Romanized Sinhala typed in Latin letters. A recent benchmark reported that language models are roughly 312 times worse at Romanized than at Sinhala-script text, and that model scale predicts nothing. Both conclusions depend on the metric. Re-measuring 31 checkpoints with tokenizer-independent bits per byte, we reproduce the published perplexities and then split the reported ratio exactly in two: on the same 24 checkpoints a factor of 21 comes from identical content being cut into a different number of tokens, and a factor of 24 from a real rise in per-byte loss. Under bits per byte, parameter count is associated with Sinhala-script ability after all. We then run the first downstream evaluation of Sinhala script variation, over three tasks, 9,479 items and eleven instruction-tuned checkpoints in both scripts, for 208,538 generations. Romanization compresses a 22.4 point spread in Sinhala-script accuracy into 6.7 points, and Romanized accuracy rises with Sinhala-script accuracy at a slope of only 0.22 (95% CI 0.11 to 0.54), so roughly four fifths of a checkpoint’s advantage does not survive the change of script. On offensive-language detection, accuracy barely moves while Matthews correlation loses more than half its value. This is not a tokenization-length artifact, because Romanized prompts are shorter in tokens for every checkpoint. Finally, absolute Sinhala-script bits per byte tracks downstream fragility, whereas the bits-per-byte *gap*, the quantity prior work reports, does not.

1 Introduction

Sinhala is an Indo-Aryan language written in a script of its own and spoken by over 17 million people, almost all in Sri Lanka (de Silva, 2019; Ranasinghe et al., 2025). Like several South Asian languages it has an everyday orthographic split.

Formal writing uses the Sinhala script, but comments, chat and social media are dominated by Romanized Sinhala, informally called Singlish, typed with Latin letters. It has no standard spelling, so කොහොමද [kəhomaḍa] (“how are you”) is typed *kohomada*, *kohomd* or *kohomade* depending on the writer.

This is not a marginal input condition. Such splits are driven by durable social and economic factors rather than fashion (Sengupta et al., 2024), and in a maternal-health service handling real user messages, Khullar et al. (2025) found 83% of Hindi and 73% of Telugu messages Romanized, with models handling them measurably worse. Anyone deploying a Sinhala system is therefore serving mostly Romanized input while evaluating on Sinhala-script benchmarks.

Rajapakse and Weerasinghe (2026) gave the first systematic measurement for Sinhala. Benchmarking 24 open-weight models with perplexity, they reported a median 312-fold degradation from Sinhala-script to Romanized text and no correlation between parameter count and script handling. Both findings are widely quotable, and both rest on a metric that is not comparable across models. Perplexity is normalised by token count, and token count for a given string depends entirely on the tokenizer (Gao et al., 2020; Magnusson et al., 2024). Sinhala is exactly where that matters, because tokenizers segment it very differently (Petrov et al., 2023; Ahia et al., 2023). That work also notes that perplexity says nothing about whether a model can still do anything useful, and calls for downstream evaluation. We take up both problems.

Fixing the metric changes the intrinsic picture.

We recompute bits per byte, per character and per word for 31 checkpoints, 7 of them new relative to prior work, over the same parallel corpus (§4). Our perplexities reproduce the published values to a median of 0.05%, so what follows comes from

the metric and not the measurement. Across all 31 checkpoints perplexity and bits per byte are unrelated ($\rho = 0.08$, $p = 0.68$), and perplexity is instead almost fully explained by tokenizer fertility ($\rho = -0.87$, $p < 10^{-9}$). Splitting the reported ratio exactly in two, on prior work’s own 24 checkpoints a factor of 21 is token-count normalisation and a factor of 24 is a real rise in per-byte loss. Under bits per byte, parameter count *is* associated with Sinhala-script ability ($\rho = -0.55$, $p = 0.001$), though still not with Romanized ability.

Romanization flattens rather than degrades. Sinhala-script cost spans 16.3 to 34.1 bits per word across checkpoints, while Romanized cost collapses into a 20.3 to 25.7 band. Twelve of the 31 are *cheaper* on Romanized text, because they were bad at Sinhala to start with, and a held-out character 4-gram model fit on 400 sentences beats all 31 on Romanized Sinhala.

The first downstream evidence for Sinhala. We evaluate eleven instruction-tuned checkpoints on 9,479 items across multiple-choice knowledge, offensive-language detection and physical commonsense, under both scripts, for 208,538 generations (§5). Every comparison is paired and tested with McNemar’s test under Holm correction. The same flattening appears: a 22.4 point spread in Sinhala-script accuracy becomes 6.7 points, and Romanized accuracy rises with Sinhala-script accuracy at a slope of only 0.22 (95% CI 0.11 to 0.54), reliably below the 1.0 of full transfer and above the 0.0 of total flattening. The same pattern holds across 16 domain by difficulty cells, and the effect is driven neither by token-length inflation nor by output degeneration.

A usable predictor, and a released transliterator. Absolute Sinhala-script bits per byte tracks the downstream gap ($\rho = -0.83$), while the bits-per-byte *gap*, the quantity prior work reports, does not ($\rho = -0.15$). Building the Romanized side needed a Sinhala-to-Latin transliterator, the opposite of the direction most Sinhala tools support. Ours beats three alternatives on 755,000 human-romanized reference items and is released as a pip-installable package.

2 Related Work

Sinhala and low-resource evaluation. Sinhala is low-resource and its tooling has grown in a fragmented way (de Silva, 2019; Ranathunga and

de Silva, 2022; Joshi et al., 2020). Recent resources are almost entirely Sinhala-script: SinhalaMMLU (Pramodya et al., 2025), SOLD (Ranasinghe et al., 2025), the Sinhala split of Global PIQA (de Silva and Ranathunga, 2026; Chang et al., 2025) and encoder work such as Dhananjaya et al. (2022). The only benchmark treating script as a variable is Rajapakse and Weerasinghe (2026), which we extend.

Romanization gaps elsewhere. Romanized South Asian text has been studied mainly as a resource and identification problem (Roark et al., 2020; Benton et al., 2025). J et al. (2024) show romanization can *help* when models are adapted for it, the mirror image of our zero-shot setting. The closest same-language work trains a model for the register: Chamikara and Sumanathilaka (2026) fine-tune multilingual encoders for hate speech detection on Romanized Sinhala, and Hewapathirana and Sumanathilaka (2024) build a classifier over Romanized Sinhala social media. We instead ask what general-purpose generative models do out of the box. Closest in spirit is Khullar et al. (2025), who find weighted F1 falling by as much as 24 points on Romanized clinical queries. We add a paired item-level design with a competence screen, a tokenizer-independent account of the same models, and a test of whether the gap is a tokenization artifact.

Why not perplexity. Bits per byte exists so that models with different vocabularies can be compared (Gao et al., 2020), and Paloma adopts it for that reason (Magnusson et al., 2024). Tokenizers treat languages very unequally (Petrov et al., 2023; Ahia et al., 2023) and over-segmentation costs accuracy on low-resource tasks (Nag et al., 2024). To our knowledge ours is the first work to show that the substantive conclusions of a published benchmark change when the normaliser changes.

3 Experimental Setup

3.1 Script conditions and data

Every item is evaluated twice, once in the Sinhala script as published and once in a Romanized form with identical content, so all comparisons are paired and differ only in orthography. This avoids the confound of an independently authored Latin-script set, of which Global PIQA does publish one. The intrinsic corpus is the parallel set released with Rajapakse and Weerasinghe (2026):

500 sentence pairs whose Romanized side was typed by native speakers, so the intrinsic results carry no dependence on our transliterator. A separate 500-sentence corpus of authentic mixed-script text supports a secondary condition.

Downstream we use every available item with no sampling. SinhalaMMLU is a 6,879-item, curriculum-aligned multiple-choice benchmark in the style of MMLU (Hendrycks et al., 2021), over six domains and three difficulty levels, of which 2,145 items have five options rather than four, so random guessing scores 23.4% rather than 25%.¹ SOLD supplies the complete 2,500-item test split of a binary offensive-language task in which 40.6% of posts are offensive, so always predicting the majority class scores 59.4%. Global PIQA, which follows PIQA (Bisk et al., 2020), contributes the 100 Sinhala items of its released non-parallel split, 77 of them culturally specific. Together the three sets give 9,479 items, or 18,958 prompts per checkpoint.

3.2 Building the Romanized side

Almost every Sinhala transliteration tool runs Latin to Sinhala (De Mel et al., 2025; Sumanathilaka et al., 2025). We need the reverse, so we implemented a deterministic grapheme-level transliterator that parses Sinhala as the abugida it is and uses the digraph conventions speakers actually type, so th is th and w is w . It runs locally, covers every input, and collapses only the consonant distinctions casual Singlish also collapses. For example $\text{වෙසක් උත්සවයේදී බෞද්ධයින්}$ [vesak uθsavaːjeːðiː bauðð^hΔjm] becomes *wesak uths-awayeedii bauddhayin*.

We validated it against 755,000 human-romanized items from the Swa-bhasha Resource Hub (Sumanathilaka et al., 2025): 4,397 authentic social-media sentence pairs, 450,587 words each carrying several accepted spellings, and a 300,000-sentence machine-augmented cross-check. Scored against the closest accepted reference by character error rate, chrF (Popović, 2015) and exact match, ours has the lowest error rate of four methods on all three corpora, 0.182, 0.121 and 0.112, and a paired Wilcoxon test separates it from Aksharamukha (Rajan, 2024), uroman (Hermjakob et al., 2018) and the web romanizer of de Silva and Ranathunga

(2026) at $p < 10^{-90}$ on the two large corpora.

The residual difference from human typing is one convention rather than phonemic error: all four methods double long vowels far more often than people do, 0.81 per word against 0.12. Word by word against the human-typed side of the parallel corpus, 44.6% of our forms match exactly and a further 27.6% match once that convention is normalised, and against the multi-reference lexicon 88.6% of our spellings are ones a human actually supplied for that word, against 84.2% for the native speakers who typed the same sentences. Appendix F gives the full comparison, and §F.3 reports what the residual costs a model, which works against us rather than for us.

3.3 Models

The intrinsic evaluation covers 31 decoder-only checkpoints from 350M to 14.7B parameters, spanning the Llama (Grattafiori et al., 2024), Qwen (Yang et al., 2024, 2025), Gemma (Gemma Team et al., 2024), Mistral, BLOOM (BigScience Workshop et al., 2023), OPT, Pythia (Biderman et al., 2023) and Phi (Abdin et al., 2024) families. Twenty-four are exactly the pool Rajapakse and Weerasinghe (2026) used, and we add three newer base checkpoints and four instruction-tuned counterparts (Appendix Table 4). The downstream evaluation needs models that follow a response format, so it uses the eleven instruction-tuned checkpoints, from 1.1B to 14.7B. Two cannot follow the format at all in the Sinhala-script condition, which §3.6 handles explicitly rather than by dropping them.

3.4 Prompting and scoring

All downstream prompts are zero-shot and greedily decoded with a 40-token limit. We picked one template from three candidates using a pilot of 3,240 generations over nine checkpoints, both scripts and 20 items per dataset, scoring by accuracy with unparseable output counted wrong. The direct template won on SinhalaMMLU, the largest set, and was the only candidate that never produced an unparseable answer there. An answer-first template scored 1.4 points higher on SOLD and 7.2 higher on Global PIQA. We applied the direct template everywhere anyway, because a template chosen per dataset would confound task with prompt, and because 20 items cannot resolve differences of that size. What a paired comparison needs is not an optimal template but one that does not favour a script, and the evidence for that comes from the main run

¹The public Hugging Face release contains a 1,851-item sample. We obtained the full set from the authors, who ask that it not be redistributed, so we release our transformation code and item identifiers rather than the items (§G).

rather than the pilot, where the unparseable rate never exceeds 0.03% in either condition for the competent cohort (§5.5). Answers are extracted by pattern match and anything unparseable is scored as wrong. All three templates and their pilot scores are in Appendix A.

3.5 Statistics

Both conditions score the same items, so we use McNemar’s exact or continuity-corrected test (McNemar, 1947; Dror et al., 2018) and report the discordant pair counts. Within each dataset we control the family-wise error rate over checkpoints with Holm’s procedure (Holm, 1979). Confidence intervals on gaps, macro-F1 differences, Matthews correlation differences and regression slopes come from 10,000 paired bootstrap resamples. For SOLD we lead with Matthews correlation (Matthews, 1975; Chicco and Jurman, 2020) because the classes are imbalanced and accuracy is close to the majority baseline. Item-level effects are also fitted as logistic regressions with checkpoint fixed effects and cluster-robust standard errors by item.

3.6 The competence screen

A model at chance in the Sinhala-script condition cannot lose anything by being given Romanized input, so a small gap for such a model is evidence of nothing, and averaging it in would understate the phenomenon. We mark a checkpoint *competent* on a task when its Sinhala-script score beats the guessing baseline at $p < 0.01$ under a Poisson-binomial null and it does not put more than 90% of its answers on one option. Six of eleven pass on SinhalaMMLU, six have a non-trivial signal on SOLD, and none pass on Global PIQA. All eleven appear in every table.

4 What the Metric Was Hiding

4.1 Metrics

For a text s we accumulate the model’s total negative log-likelihood $NLL(s)$ in nats and divide by a unit of the text itself rather than by its tokenization:

$$BPB = \frac{\sum_s NLL(s)}{\ln 2 \sum_s |s|_{\text{bytes}}}, \quad (1)$$

and likewise per character (BPC) and per whitespace word (BPW). The denominator no longer depends on the tokenizer. The numerator still does,

because a model’s loss reflects how its own vocabulary carves up the string, so Equation 1 removes the arbitrary unit of account rather than every trace of tokenization. We pool numerator and denominator over the corpus rather than averaging per-sentence ratios, so short sentences do not dominate. Bytes are the standard choice (Gao et al., 2020), but BPW is sharpest across scripts, because a sentence and its romanization contain the same number of words while Sinhala UTF-8 spends 2.65 bytes per character against 1.00 for Latin.

Our recomputed perplexities reproduce the published values for all 24 shared checkpoints, with median absolute deviation 0.05% on Sinhala-script and 0.004% on Romanized text, and 0.39% on mixed script once two outliers are excluded. Appendix Table 5 sets our recomputation beside every headline number of the original benchmark, so everything below reflects the metric and not a different run.

4.2 Perplexity ranks models by their tokenizers

Figure 2 makes the central methodological point. Across the 31 checkpoints, Sinhala-script perplexity and Sinhala-script bits per byte are statistically unrelated ($\rho = 0.08$, $p = 0.68$), and so are the Romanized values ($\rho = 0.13$, $p = 0.48$). Only one of the five best by perplexity is among the five best by bits per byte. Gemma-2-9B ranks 26th of 31 on perplexity and 2nd on bits per byte, and Qwen3.5-9B-Base moves from 28th to 6th.

The cause is visible in Figure 2(b). Perplexity is almost fully explained by how many tokens a tokenizer spends on a Sinhala word ($\rho = -0.87$, $p < 10^{-9}$), which ranges from 4.3 for the Qwen3.5 and Gemma tokenizers to 14.2 for models that fall back to bytes. Spreading the same loss over three times as many tokens lowers per-token perplexity without the model knowing anything more, so a headline perplexity comparison across families on Sinhala is largely a comparison of tokenizers.

How completely the two metrics come apart depends on the pool. On prior work’s 24 checkpoints the rank correlation is moderate ($\rho = 0.46$, $p = 0.02$), still leaving four fifths of bits-per-byte rank variance unexplained, and adding the seven newer checkpoints removes even that. The disagreement is therefore not an artifact of our additions, but it does grow as modern tokenizers enter the comparison.

4.3 Splitting the reported ratio in two

The degradation ratio can be decomposed rather than argued about. Since perplexity is $\exp(\text{NLL}/T)$ for T scored tokens and $\text{NLL} = \ln 2 \cdot A \cdot B$ for A bits per byte over B bytes,

$$\log_2 \text{PPL} = A \cdot x, \quad x = B/T, \quad (2)$$

where x is the bytes each token carries. Across the two script conditions the change in $\log_2 \text{PPL}$ is then exactly

$$\underbrace{\bar{A}(x_r - x_u)}_{\text{token-count normaliser}} + \underbrace{\bar{x}(A_r - A_u)}_{\text{per-byte loss}}, \quad (3)$$

with $\bar{A}$ and $\bar{x}$ the two-condition means. The first term is the part of the reported degradation caused purely by identical content being cut into a different number of tokens. The second is the part caused by the model genuinely assigning less probability per byte. The split is an identity, not a model, and it holds to 5×10^{-15} bits in our data.

On prior work’s 24 checkpoints the median gap is 8.29 bits, which is the 312-fold ratio they report. Equation 3 assigns 4.37 bits to the normaliser and 4.56 to per-byte loss, a factor of 21 and a factor of 24 whose product is the published 312, so almost exactly half the reported degradation is the unit of account. Over all 31 checkpoints the normaliser share falls to 39%, and for the six Qwen3.5 and Gemma checkpoints it is slightly negative (Appendix Table 6). Those six also report the five smallest ratios in the pool, with the four Qwen3.5 entries between 58-fold and 64-fold against a pool median of 294-fold, so the eye-catching part of a 312-fold figure is concentrated in the models whose tokenizers handle Sinhala worst.

Two conclusions follow. First, the reported ratio substantially overstates how much worse these models are at Romanized Sinhala. The same measurement stated tokenizer-independently is 2.14 bits per byte, a factor of 4.4 per byte, or 1.11 bits per word. Second, the finding that scale does not predict script handling does not survive. Parameter count is uncorrelated with Sinhala-script perplexity ($\rho = 0.15$, $p = 0.42$) but correlated with Sinhala-script bits per byte ($\rho = -0.55$, $p = 0.001$) and with mixed-script bits per byte ($\rho = -0.57$, $p = 0.0008$), and the association holds on prior work’s own 24 checkpoints ($\rho = -0.42$, $p = 0.04$). What scale does not buy is Romanized ability ($\rho = -0.28$, $p = 0.13$). Since these checkpoints

differ in family, tokenizer and training corpus at once, this is a cross-model association and not a controlled scaling result.

4.4 Romanization flattens the field

Figure 1(a) shows the effect that motivates the rest of the paper. Sinhala-script cost spans 16.3 to 34.1 bits per word, a spread of 17.7 bits, while Romanized cost spans 20.3 to 25.7, a spread of 5.4. Regressing one on the other, a full bit per word of Sinhala-script improvement buys only 0.14 bits on Romanized text (95% CI 0.05 to 0.25, $R^2 = 0.19$). Twelve of the 31 checkpoints are cheaper on Romanized text, and which ones is predictable: the better a checkpoint is on the Sinhala script, the more it loses ($\rho = -0.88$, $p < 10^{-9}$). Romanized Sinhala is a regime in which model quality has largely stopped mattering. The mixed-script condition behaves quite differently and reproduces the earlier finding, with Sinhala-script and mixed bits per byte strongly rank-correlated ($\rho = 0.94$) at a median gap of only 0.49 bits, against $\rho = 0.41$ for Romanized. Interleaving scripts is a mild perturbation. Replacing one entirely is not.

4.5 How much is that in absolute terms?

Relative statements leave open whether models are merely worse or genuinely uninformed. As a reference we fit add-1 character n -gram models on the same corpus under five-fold cross-validation, so each score is out of sample. On Romanized Sinhala the best n -gram reaches 2.64 bits per byte and a character bigram 3.06, while the best of 31 checkpoints reaches 3.29 and the median 3.81, so not one beats the bigram. On the Sinhala script the picture is ordinary, with the character trigram at 1.16 and one checkpoint below it (Appendix Table 8). The n -gram is a cheap in-domain model and the comparison favours it in that respect, so we read this as Romanized-condition checkpoints failing to exploit structure beyond what a character bigram captures, rather than as a claim about what they know.

5 Downstream Consequences

Table 2 reports every checkpoint on SinhalaMMLU and SOLD.

5.1 Multiple-choice knowledge

Four of the six competent checkpoints lose significantly under Romanization after Holm correction. Qwen3.5-9B falls from 44.1% to 28.7%, a 15.5

Checkpoint	Sinhala			Rom.			Checkpoint	Sinhala			Rom.		
	B	PPL	BPB	BPW	BPB	BPW		B	PPL	BPB	BPW	BPB	BPW
Llama-3.1-8B	8.0	3.22	1.148	16.3	3.584	22.1	Cerebras-GPT-1.3B	1.3	2.86	1.491	21.2	3.700	22.8
Gemma-2-9B	9.2	13.23	1.190	16.9	3.999	24.7	Mistral-7B-v0.3	7.2	5.26	1.559	22.2	3.884	24.0
Llama-3.2-3B	3.2	3.74	1.296	18.4	3.831	23.6	Hormoz-8B‡	8.0	5.98	1.567	22.3	4.045	25.0
Llama-3.1-8B-Instruct‡	8.0	3.86	1.326	18.9	3.718	22.9	Minitron-8B-Base	8.4	2.98	1.575	22.4	3.428	21.2
Phi-4-14B‡	14.7	4.01	1.341	19.1	3.446	21.3	Qwen1.5-1.8B	1.8	8.07	1.601	22.8	3.573	22.1
Qwen3.5-9B-Base	9.0	21.39	1.350	19.2	3.290	20.3	BLOOM-3B	3.0	6.95	1.748	24.8	3.626	22.4
Pythia-410M	0.4	2.58	1.351	19.2	3.707	22.9	OPT-2.7B	2.7	3.59	1.833	26.1	3.974	24.5
Qwen3.5-9B‡	9.0	22.56	1.373	19.5	3.311	20.4	SmolLM3-3B‡	3.1	6.98	1.876	26.7	4.145	25.6
Llama-3.2-1B	1.2	4.11	1.387	19.7	3.881	24.0	OPT-1.3B	1.3	3.87	1.942	27.6	4.010	24.8
Qwen3.5-4B-Base	4.0	23.92	1.399	19.9	3.378	20.8	Zephyr-7B-beta‡	7.2	8.28	1.986	28.2	4.167	25.7
Gemma-7B	8.5	21.13	1.406	20.0	3.805	23.5	BLOOM-1B1	1.1	11.00	2.162	30.7	3.666	22.6
Qwen2-7B	7.6	6.29	1.410	20.0	3.498	21.6	StableLM-Zephyr-3B‡	2.8	4.57	2.163	30.8	3.986	24.6
Qwen2-7B-Instruct‡	7.6	6.46	1.431	20.3	3.581	22.1	OPT-350M	0.3	4.75	2.238	31.8	3.986	24.6
Qwen3.5-4B‡	4.0	25.84	1.433	20.4	3.422	21.1	BLOOM-560M	0.6	12.64	2.287	32.5	3.810	23.5
TinyLlama-1.1B-Chat‡	1.1	4.38	1.469	20.9	3.923	24.2	LaMini-GPT-1.5B‡	1.5	5.41	2.395	34.1	3.827	23.6
Mistral-Nemo-Base-2407	12.2	2.80	1.486	21.1	3.896	24.0							

Table 1: All 31 checkpoints, sorted by Sinhala-script bits per byte and continued in the right-hand block. ‡ marks the eleven evaluated downstream. The two orderings disagree almost completely: Gemma-2-9B is 2nd by bits per byte and 26th by perplexity. Remaining columns are in Appendix Table 7.

point drop (95% CI 14.0 to 16.9, 1,892 against 828 discordant pairs), Qwen3.5-4B loses 10.6 points, Llama-3.1-8B-Instruct 5.4 and Qwen2-7B-Instruct 2.0. The other two, Hormoz-8B and Phi-4-14B, have only 2.5 and 5.2 points of headroom and lose less than 1.5 points each, neither significantly. Pooled over the cohort the gap is 6.0 points (95% CI 5.4 to 6.5, $n = 41,274$ paired items), and a logistic fit with checkpoint fixed effects and item-clustered errors puts the odds ratio for Romanized input at 0.70 ($p < 10^{-33}$).

Read as levels the result is starker. Figure 1(b) shows Sinhala-script accuracy spanning 22.4 points while Romanized accuracy spans 6.7, and Romanized accuracy never exceeds 28.7% against a 23.4% chance baseline, so the best checkpoint keeps 5.2 of its 20.7 points of headroom. Five of the seven checkpoints with no significant gap are already at or near chance natively.

5.2 How much Sinhala-script ability survives

The natural way to state the relationship is that the loss grows with the headroom a checkpoint had, and that framing is a trap worth flagging. Writing headroom as Sinhala-script accuracy minus chance, a fit across the ten parseable checkpoints gives $\text{loss} = 0.78 \times \text{headroom} - 1.06$ with $R^2 = 0.96$. But Sinhala-script accuracy appears on both axes, so a high fit is close to guaranteed: shuffling the Romanized scores to destroy any association still yields a mean slope of 1.00 and a mean R^2 of 0.94, with 8.9% of permutations reaching $R^2 \geq 0.96$. We do not treat that fit as evidence.

The same measurements stated as levels do carry information. Regressing Romanized on Sinhala-

script accuracy gives a slope of 0.22 (95% bootstrap CI 0.11 to 0.54, leave-one-checkpoint-out range 0.17 to 0.26, $R^2 = 0.65$), reliably below the 1.0 of full transfer ($p = 8.6 \times 10^{-7}$) and above the 0.0 of complete flattening, since only 0.5% of permutations reach the observed fit. Romanization removes roughly four fifths of whatever Sinhala-script advantage a checkpoint had, and the leave-one-out range shows this is not the work of one or two influential points.

Ten checkpoints is thin, so we test the same relation inside the benchmark, where the units are items rather than models. Splitting SinhalaMMLU into its 16 non-empty domain by difficulty cells and pooling the competent cohort within each gives a slope of 0.45 (95% CI 0.36 to 0.63, leave-one-cell-out 0.43 to 0.50, $R^2 = 0.75$), again below 1.0 at $p = 1.5 \times 10^{-6}$, and 0.48 when cells are weighted by item count. The two views are complementary rather than independent, since they aggregate the same generations differently, but they agree on the shape and disagree usefully on the amount: content-level transfer is about twice model-level transfer (Appendix Table 12).

The strata make the same point (Appendix Table 9). Loss falls monotonically with difficulty, 7.9 points on Easy items, 6.2 on Medium and 4.3 on Hard, with a significant interaction in the item-clustered logistic fit ($p < 0.001$), because models are nearer chance on hard items in both conditions. By domain it runs from 10.3 points on Social Science, where Sinhala-script accuracy is highest, to 4.1 on Humanities, the largest domain. The harder and less well-served a slice is, the smaller the script effect it can reveal.

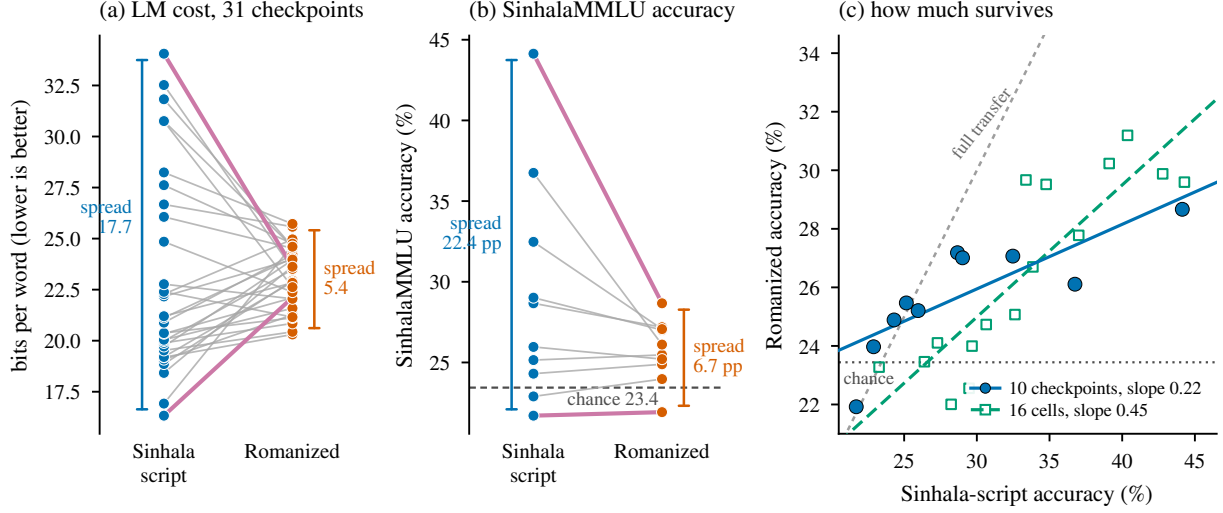


Figure 1: Romanization flattens the field. (a) Language modelling cost for 31 checkpoints. Word counts are identical within a pair, so the columns are directly comparable, and the highlighted lines are checkpoints cheaper on Romanized text. (b) SinhalaMMLU accuracy: a 22.4 point spread becomes 6.7 and no checkpoint finishes more than 5.3 points above chance. (c) The same numbers as levels, with the 16 domain by difficulty cells. Both slopes fall well below the full-transfer slope of 1 (bootstrap intervals 0.11 to 0.54 and 0.36 to 0.63). LaMini-GPT-1.5B is omitted from (b) and (c) for unparseable output on 95.9% of Sinhala-script items.

Checkpoint	B	SinhalaMMLU (6,879 items)						SOLD (2,500 items)				
		Acc. %			Δ	95% CI	p_{Holm}	macro-F1		MCC		p_{Holm}
		S	R					S	R	S	R	
TinyLlama-1.1B-Chat	1.1	21.7	21.9	-0.2	[-0.5, 0.1]	.8273	6.4	24.2	-0.019	-0.043	1.0000	
LaMini-GPT-1.5B	1.5	0.9	18.8	-17.9	[-18.8, -16.9]	< .0001	5.1	28.3	0.000	0.024	1.0000	
StableLM-Zephyr-3B	2.8	22.9	24.0	-1.1	[-2.3, 0.1]	.4275	38.6	38.9	0.036	0.001	.9408	
SmolLM3-3B	3.1	25.1	25.5	-0.3	[-1.0, 0.4]	.8273	29.7	29.9	0.025	0.017	1.0000	
Qwen3.5-4B†	4.0	36.7	26.1	+10.6	[9.3, 12.0]	< .0001	60.2	55.1	0.219	0.127	.0018	
Zephyr-7B-beta	7.2	24.3	24.9	-0.6	[-1.7, 0.5]	.8273	39.7	39.6	0.060	0.060	1.0000	
Qwen2-7B-Instruct†	7.6	29.0	27.0	+2.0	[0.8, 3.2]	.0055	49.9	50.3	0.148	0.042	.0018	
Llama-3.1-8B-Instruct†	8.0	32.5	27.1	+5.4	[4.1, 6.7]	< .0001	60.1	52.6	0.259	0.105	< .0001	
Hormoz-8B†	8.0	26.0	25.2	+0.8	[-0.5, 2.0]	.8273	47.9	51.9	0.108	0.090	1.0000	
Qwen3.5-9B†	9.0	44.1	28.7	+15.5	[14.0, 16.9]	< .0001	63.1	52.0	0.280	0.069	< .0001	
Phi-4-14B†	14.7	28.7	27.2	+1.5	[0.2, 2.8]	.1286	54.4	50.8	0.152	0.076	.0084	

Table 2: Downstream results, ordered by size. S is the Sinhala script, R is Romanized, Δ is the paired difference in points, p_{Holm} is McNemar’s test corrected across checkpoints within a task, and MCC is Matthews correlation with a paired-bootstrap p . † marks checkpoints passing the competence screen. Global PIQA and the remaining columns are in Appendix E.

5.3 Offensive language: accuracy hides the damage

SOLD is where the choice of metric decides the conclusion. Accuracy falls by 1.6 to 11.3 points against a 59.4% majority baseline, which looks survivable. Matthews correlation says otherwise (Appendix Figure 3): Qwen3.5-9B goes from 0.280 to 0.069, keeping 24% of its signal, Qwen2-7B-Instruct from 0.148 to 0.042, keeping 29%, and Llama-3.1-8B-Instruct from 0.259 to 0.105, keeping 40%. Five of the six checkpoints with a real Sinhala-script signal lose significantly under Holm correction and the median keeps 45%. The mecha-

nism is a retreat toward trivial behaviour: Qwen2-7B-Instruct more than doubles its rate of predicting the offensive class, 10.5% to 23.4%, without becoming more accurate. A deployment tracking accuracy would see a five point wobble while over half the usable signal disappeared.

5.4 Global PIQA is underpowered, and says so

No checkpoint is above chance in the Sinhala-script condition of Global PIQA, so the screen admits none, and with 100 items and a median of 36 discordant pairs no comparison is significant. The direction is consistent, with seven of ten parseable check-

points dropping, but a sign test gives $p = 0.18$ and pooling gives a 3.8 point gap at $p = 0.14$. We report this as a null result rather than corroboration: a 100-item set cannot resolve effects of the size we measure elsewhere, which is worth saying given how often small multilingual splits are read as evidence.

5.5 Two things it is not

Two mechanisms can be ruled out. The first is token fragmentation. Romanization does not lengthen the input for Sinhala: the Sinhala script costs a median 9.5 tokens per word against 2.09 for its romanization, because these tokenizers largely fall back to bytes, and all eleven checkpoints receive *fewer* tokens in the Romanized condition on all three tasks, a median ratio of 0.48 on SinhalaMMLU. Models get shorter, more compactly encoded input and do worse, and splitting items into terciles by how much romanization inflates tokens gives no monotone pattern either, with losses of 5.1, 6.6 and 6.2 points. This is inconsistent with token-length inflation as the primary explanation and consistent with insufficient exposure to Romanized Sinhala in pretraining, though it does not establish it, since we cannot audit these corpora.

The second is output degeneration, since a model could score at chance by emitting one constant answer. For the competent cohort the share of answers on the most frequent option is 37.3% on the Sinhala script and 39.8% under romanization (Wilcoxon $p = 0.31$), and the invalid rate never exceeds 0.03% in either condition. The models keep answering across the option set. They are simply wrong. Appendix E gives both controls per checkpoint.

6 Do Intrinsic Metrics Track Downstream Fragility?

Can a cheap intrinsic measurement stand in for an expensive downstream one? For the ten checkpoints measured both ways, absolute Sinhala-script bits per byte tracks the SinhalaMMLU gap ($\rho = -0.83$, 95% CI -1.00 to -0.30 , leave-one-out -0.90 to -0.77) and the loss of Matthews correlation on SOLD ($\rho = -0.78$, CI -1.00 to -0.36 , leave-one-out -0.87 to -0.73), which is the flattening of §5.2 seen from the intrinsic side. The negative result is the more useful one: the bits-per-byte *gap*, the natural analogue of the degradation ratio prior work reports, shows no association

($\rho = -0.15$, CI -0.89 to $+0.63$), and neither does the perplexity ratio ($\rho = -0.52$). For the gap even the leave-one-out range straddles zero, so we cannot sign it. With ten checkpoints that is an absence of evidence rather than evidence of absence, but the widely reported quantity gave us nothing to act on while the level did.

All of this is exploratory: ten heterogeneous checkpoints, no held-out validation, no pre-registration, and intervals wide enough to reach -1.00 . The levels-versus-gaps contrast is stable under leave-one-out, but this is a hypothesis for a larger checkpoint set rather than an instrument (Appendix Table 17).

7 Discussion

Three things follow for practice. First, a benchmark reporting perplexity across model families on a script that tokenizers handle unevenly is reporting tokenizer behaviour as much as model behaviour, and bits per byte costs nothing extra on the same forward passes. Second, script robustness cannot be read off a group average, because checkpoints at chance contribute a zero gap by construction and the hardest strata reveal the smallest effects, so any claim needs a competence screen and a stated baseline. Third, the checkpoints best on the Sinhala script have the most to lose, so picking by Sinhala-script scores and serving Romanized users selects for the largest drop. Since Romanized ability is nearly flat across a 42-fold parameter range, the remedy is transliteration before inference and Romanized pretraining data.

Limitations

One language. Everything here is Sinhala. The mechanism we describe, a floor effect that makes robustness gaps proportional to Sinhala-script ability, should generalise to any language with a dominant Romanized register, but we have not shown that. Hindi, Nepali, Telugu, Urdu, Tamil and Arabic dialects are the obvious tests, and Khullar et al. (2025) report compatible magnitudes for Hindi, Nepali, Telugu, Kannada, Marathi and Punjabi.

The Romanized side of the downstream sets is synthetic, and it is the harder end of the range. We transliterate Sinhala-script items rather than collecting human-typed Romanized versions of them, because no such parallel resource exists at this scale. We measured what that substitution costs rather than assuming it. On the 500 sentences of the

parallel corpus, which has a Romanized side that native speakers typed, we also scored our transliterator’s output for the same sentences. Our text costs 0.17 to 0.24 more bits per byte than what the humans wrote, positive for all six checkpoints we could rerun (§F.3). Our Romanized condition is therefore somewhat harder than authentic typing, which makes the downstream gaps we report closer to upper bounds than to lower bounds on what a user would experience. This cuts against the direction we would have guessed from the spelling statistics alone, and we report it because it bounds the claim rather than because it helps it.

Two things limit how far that control carries. It covers six of 31 checkpoints, all below 3.2B parameters, and none of the six passes the downstream competence screen. For the checkpoints that do carry the downstream result we instead test the same worry item by item on SOLD, and find that how closely an item’s Romanized spellings match attested human spellings does not predict its Romanized penalty, with a Spearman correlation of 0.02 ($p = 0.31$, $n = 2,447$) in Table 18. The intrinsic results do not share this limitation at all, since their Romanized side was typed by native speakers, and they show the same flattening. A human-written Romanized version of even a few hundred SinhalaMMLU items would still settle the question better than any of this.

Checkpoint size is confounded with everything else. Our 31 checkpoints differ in family, tokenizer, training corpus and instruction-tuning status at the same time. The association between parameter count and Sinhala-script bits per byte should not be read as a scaling law, and our own fertility analysis is direct evidence that family-level confounds are present.

We cannot exclude contamination. The training corpora of these checkpoints are not fully auditable, so we cannot rule out that public evaluation sets, or Sinhala-script material closely related to them, appeared during pretraining. If they did, the Sinhala-script side would be inflated and the gaps we report would be overstated. This applies to any evaluation of pretrained models on public benchmarks and we have no way to bound it here.

The intrinsic-to-downstream link is exploratory. Ten checkpoints, no held-out validation, no pre-registration, and bootstrap intervals wide enough to reach -1.00 (§6). Treat Table 17 as a hypothesis,

not as a screening tool.

The mixed-script condition is weaker evidence than the rest. That corpus is authentic mixed-script text rather than a transliteration of the parallel set, so it is not parallel to either script condition. It supports a descriptive statement about how mildly script mixing perturbs cost, and nothing paired.

Open weights only, and small ones. We evaluate open checkpoints up to 14.7B for the downstream tasks. Frontier proprietary models are much stronger on Sinhala (Pramodya et al., 2025) and therefore have far more headroom, so the flattening slope of §5.2 predicts larger absolute losses for them. That prediction is untested here.

Zero-shot only. One template, zero-shot, greedy. This measures what a user gets out of the box, not the ceiling. Few-shot prompting, transliterate-then-infer, or Romanized fine-tuning could all narrow the gap, and we do not measure by how much. The template was chosen on a 20-item pilot per dataset, and 20 of those SinhalaMMLU items, 0.3% of the benchmark, are also in the final evaluation set.

Two checkpoints cannot do the task. LaMini-GPT-1.5B and TinyLlama-1.1B-Chat produce unparseable output on most Sinhala-script items of at least one task. Their apparent improvement under Romanization is an artifact of that failure, not a script effect, which is why the competence screen exists.

Ethics Statement

Content warning. §5 and Appendix E discuss an offensive-language dataset. We quote no offensive text anywhere in this paper, and report only aggregate metrics over it.

Data. All datasets are public research releases used for evaluation only, and Appendix G states each licence and exactly what we do and do not redistribute. Three points matter. Global PIQA forbids training on it or on synthetic data seeded from it, and we do neither. SinhalaMMLU is released under a licence that forbids distributing adaptations, and its authors asked that the full set not be made public, so our Romanized version of it stays local and we release only the code and the item identifiers needed to rebuild it. SOLD contains offensive social media text. It is already pseudonymised upstream, with user mentions replaced by placeholders, and we neither collected nor re-annotated

any of it. We add no new human annotation and no new data collection, so no ethics approval was required.

Models. All 31 checkpoints were used for evaluation only, with no fine-tuning, redistribution or derivative model release. That is within the terms of each family’s licence, including the Llama Community License, the Gemma Terms of Use and the BigScience RAIL licence, all of which permit evaluation of the released weights.

Risk. The practical risk our results speak to is deployment. A moderation or support system validated on Sinhala-script text and serving Romanized users will look far better in evaluation than in the field, and on SOLD that failure is invisible in accuracy while more than half the usable signal is gone. Publishing the size of that gap, and a screen that stops averaging from hiding it, is the useful response. Our transliterator collapses several Sinhala consonant distinctions, matching casual Singlish practice, so it is a tool for building evaluation data and not for rendering names or other text where those distinctions carry meaning.

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A Prompt Templates

One template, T1 below, is used for every dataset and both script conditions. T2 and T3 were the pilot alternatives (§3.4) and are given here for completeness. {n} is the item’s option count and {domain} is the domain name where the dataset provides one.

T1, direct (used for every result in this paper).

This is a multiple-choice question related to the {domain}. Choose the correct or most appropriate answer from answers 1/{...}/{n} for the following question. Do not repeat the question or text in your response. Respond with only the number (1/{...}/{n}) and nothing else.
Question: {question}
1. {option 1} ... {n}. {option n}
Answer:

Template	Accuracy %			Invalid %	
	S	R	both	S	R
<i>SinhalaMMLU</i>					
T1 direct (used)	28.3	25.0	26.7	0.0	0.0
T2 two-shot	18.3	28.9	23.6	10.6	2.8
T3 answer first	27.2	23.9	25.6	0.6	0.6
<i>SOLD</i>					
T1 direct (used)	51.1	51.1	51.1	10.6	8.9
T2 two-shot	42.8	44.4	43.6	10.6	5.6
T3 answer first	49.4	55.6	52.5	11.1	8.9
<i>Global PIQA</i>					
T1 direct (used)	48.9	40.6	44.7	4.4	3.3
T2 two-shot	46.1	48.3	47.2	13.9	0.6
T3 answer first	52.2	51.7	51.9	3.9	5.6

Table 3: The prompt-template pilot: 9 checkpoints, 3 templates, 2 script conditions, 20 items per dataset, 3,240 generations. Accuracy counts unparseable output as wrong. T1 was applied to all three datasets, although T3 scored higher on SOLD and Global PIQA, so that no dataset gets a different prompting regime (§3.4). At 20 items per dataset none of these differences is resolvable, which is why the pilot is treated as a format check.

For Global PIQA, which has no domain field, the first sentence is replaced by “Out of the given {n} answers, which one is the best option to complete the given text?”. Binary classification:

Classify the following Sinhala text as either ‘NOT’ (not offensive) or ‘OFF’ (offensive). Do not repeat the question or text in your response. Respond with only ‘NOT’ or ‘OFF’ and nothing else.
Text: {text}
Answer:

T2, two-shot. As T1, with “exactly like the examples below” appended to the instruction and two fixed exemplars inserted before the item. The exemplars were authored separately and appear in no evaluation set.

T3, answer first. As T1, with the final instruction replaced by “State the answer number (1/{...}/{n}) FIRST, before any explanation. You may briefly explain your reasoning only after giving the number.”

Prompts are rendered through each checkpoint’s own chat template. For checkpoints that emit reasoning traces by default we disable that behaviour identically in both script conditions. Decoding is greedy at temperature 0 with at most 40 new tokens and inputs truncated at 4,096 tokens.

B Checkpoints

Table 4 lists all 31 checkpoints with their Hugging Face identifiers and their tokenizer fertility on the

parallel corpus. Note the consistency of Romanized fertility, between 1.9 and 2.4 tokens per word, against a range of 4.3 to 14.2 on the Sinhala script.

C Reproducing the Published Benchmark

Table 5 sets our recomputation beside every headline number of Rajapakse and Weerasinghe (2026), on exactly the 24 checkpoints they evaluated. Our median degradation ratio is 312.3 against their published 312.3, and per-checkpoint perplexities agree to a median of 0.05%. Two mixed-script values, for Phi-4-14B and Mistral-Nemo-Base-2407, differ by more than 5% and are the only points where we cannot reproduce their run. Our conclusions differ from theirs because of the metric, not because of the measurement.

The two parameter-count correlations flip sign between the two runs, from +0.09 to -0.05 and from -0.09 to +0.11. Both are within noise of zero on 24 points and neither is significant in either run, so the reproduction agrees on what matters, which is that perplexity gives no signal on scale.

D Full Intrinsic Results

Table 7 gives every metric for every checkpoint in all three script conditions, corpus-pooled as in Equation 1. Table 8 and Figure 4 give the character n -gram reference of §4.

E Additional Downstream Detail

Table 10 gives the unparseable-output rates and the input token ratios omitted from Table 2, Table 9 the SinhalaMMLU strata, and Table 11 the per-checkpoint Global PIQA results.

One checkpoint lets us check the pretraining-exposure reading of §5.5 directly. BLOOM’s corpus covers 46 languages and Sinhala is not one of them (Laurençon et al., 2023; BigScience Workshop et al., 2023), and all three BLOOM checkpoints sit at or near the bottom of Table 1 on Sinhala-script text while being unremarkable on Romanized text. Exposure is documented for no other checkpoint in the pool.

F The Transliterator

F.1 Quality against human references

Table 13 compares our method with Aksharamukha (Rajan, 2024), uroman (Hermjakob et al., 2018) and the web romanizer released with de Silva

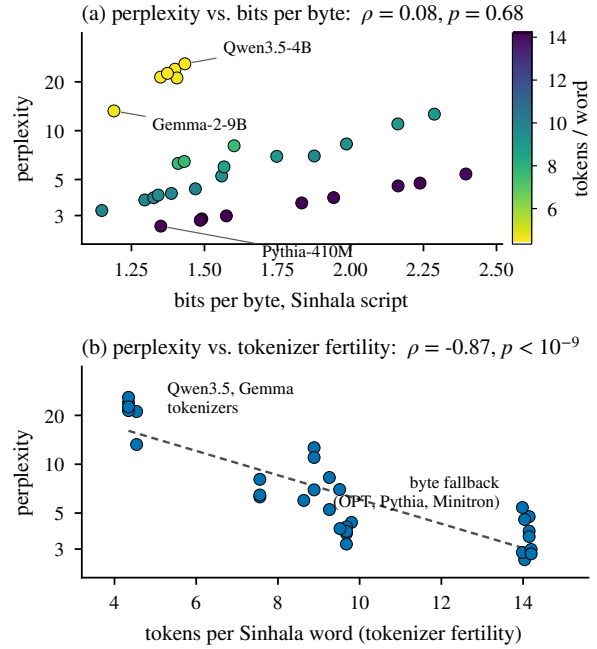


Figure 2: Perplexity does not measure what it appears to. (a) Across 31 checkpoints, Sinhala-script perplexity is unrelated to Sinhala-script bits per byte. (b) It is instead almost fully determined by how many tokens the tokenizer spends per Sinhala word.

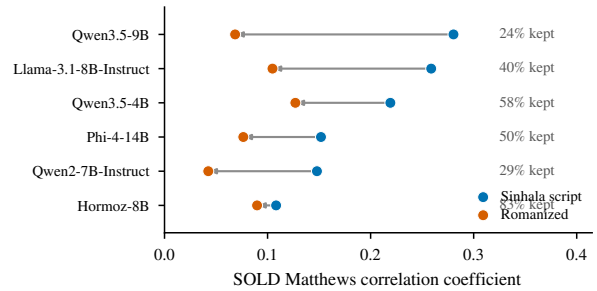


Figure 3: Matthews correlation on SOLD for the six checkpoints with a real Sinhala-script signal, with the percentage retained under Romanization. Accuracy for the same checkpoints moves by only 1.6 to 11.3 points (Table 2).

and Ranathunga (2026), against 755,000 human-romanized items. Ours has the lowest character error rate on all three corpora. The web tool is close once its v is rewritten to w , which is a spelling convention rather than a transliteration error, but it cannot romanize 17 sequences involving $ආ$ [ɹɹ] at all and leaves Sinhala characters in 0.8% to 2.5% of its output.

F.2 How our spellings differ from human typing

Aligning our output word by word against the human-typed Romanized side of the parallel cor-

Checkpoint	B params	Hugging Face identifier	tokens / word	
			Sinhala	Romanized
OPT-350M●	0.35	facebook/opt-350m	14.1	2.4
Pythia-410M●	0.41	EleutherAI/pythia-410m	14.0	2.2
BLOOM-560M●	0.56	bigscience/bloom-560m	8.9	1.9
BLOOM-1B1●	1.10	bigscience/bloom-1b1	8.9	1.9
TinyLlama-1.1B-Chat‡●	1.10	TinyLlama/TinyLlama-1.1B-Chat-v1.0	9.8	2.4
Llama-3.2-1B●	1.20	meta-llama/Llama-3.2-1B	9.7	2.2
Cerebras-GPT-1.3B●	1.30	cerebras/Cerebras-GPT-1.3B	14.0	2.3
OPT-1.3B●	1.30	facebook/opt-1.3b	14.1	2.4
LaMini-GPT-1.5B‡●	1.50	MBZUAI/LaMini-GPT-1.5B	14.0	2.3
Qwen1.5-1.8B●	1.80	Qwen/Qwen1.5-1.8B	7.6	2.1
OPT-2.7B●	2.70	facebook/opt-2.7b	14.1	2.4
StableLM-Zephyr-3B‡●	2.80	stabilityai/stablelm-zephyr-3b	14.0	2.2
BLOOM-3B●	3.00	bigscience/bloom-3b	8.9	1.9
SmolLM3-3B‡●	3.10	HuggingFaceTB/SmolLM3-3B	9.5	2.1
Llama-3.2-3B●	3.20	meta-llama/Llama-3.2-3B	9.7	2.2
Qwen3.5-4B-Base	4.00	Qwen/Qwen3.5-4B-Base	4.3	2.0
Qwen3.5-4B‡	4.00	Qwen/Qwen3.5-4B	4.3	2.0
Mistral-7B-v0.3●	7.20	mistralai/Mistral-7B-v0.3	9.3	2.4
Zephyr-7B-beta‡●	7.20	HuggingFaceH4/zephyr-7b-beta	9.3	2.4
Qwen2-7B●	7.60	Qwen/Qwen2-7B	7.6	2.1
Qwen2-7B-Instruct‡	7.60	Qwen/Qwen2-7B-Instruct	7.6	2.1
Llama-3.1-8B●	8.00	meta-llama/Meta-Llama-3.1-8B	9.7	2.2
Llama-3.1-8B-Instruct‡	8.00	meta-llama/Llama-3.1-8B-Instruct	9.7	2.2
Hormoz-8B‡●	8.00	mann-e/Hormoz-8B	8.6	2.1
Minitron-8B-Base●	8.40	nvidia/Minitron-8B-Base	14.2	1.9
Gemma-7B●	8.50	google/gemma-7b	4.5	2.0
Qwen3.5-9B-Base	9.00	Qwen/Qwen3.5-9B-Base	4.3	2.0
Qwen3.5-9B‡	9.00	Qwen/Qwen3.5-9B	4.3	2.0
Gemma-2-9B	9.20	google/gemma-2-9b	4.5	2.0
Mistral-Nemo-Base-2407●	12.20	mistralai/Mistral-Nemo-Base-2407	14.2	2.2
Phi-4-14B‡●	14.70	microsoft/phi-4	9.5	2.1

Table 4: All evaluated checkpoints, with Hugging Face identifiers and tokenizer fertility on the parallel corpus. ‡ marks the eleven used for the downstream evaluation and ● the 24 that [Rajapakse and Weerasinghe \(2026\)](#) evaluated. Sinhala-script fertility varies by more than a factor of three across tokenizers, whereas Romanized fertility is nearly constant.

Quantity	Published	Ours
median PPL ratio, Sinh. to Rom.	312.3	312.3
its range	149.1 to 769.7	148.5 to 769.3
median PPL ratio, Sinh. to mixed	2.0	2.0
ρ (params, Sinh. PPL)	+0.09	-0.05
ρ (params, Rom. PPL)	-0.09	+0.11
ρ (Sinh. PPL, Rom. PPL)	+0.52	+0.53
ρ (Sinh. PPL, mixed PPL)	+0.96	+0.98
CV, Sinh. PPL	66.4%	67.8%
CV, Rom. PPL	62.3%	63.8%
median dev. , Sinh. PPL	—	0.047%
median dev. , Rom. PPL	—	0.004%

Table 5: Our recomputation against the published benchmark, on its own 24 checkpoints. Deviations are per-checkpoint absolute percentage differences from the published perplexities.

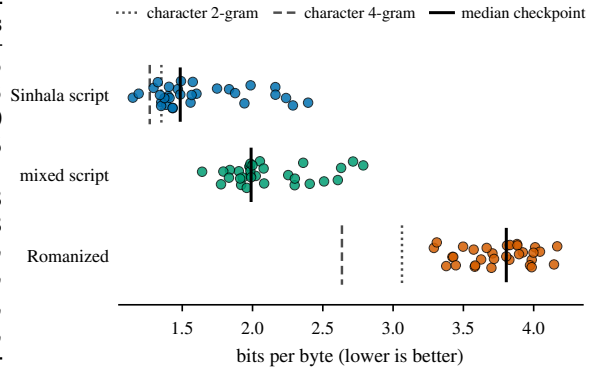


Figure 4: Bits per byte for all 31 checkpoints in each script condition, with held-out character n -gram references. The mixed-script corpus is not parallel, so no reference is shown for it.

pus, 485 of the 500 sentences have matching token counts, giving 2,933 aligned word pairs. Of those, 44.6% match exactly. A further 27.6% match once runs of the same Latin vowel are collapsed, which is the one convention our transliterator applies uni-

formly and people apply erratically. The remaining 27.8% differ in some other way, and Table 15 lists the most frequent. They are dominated by vowel reduction and abbreviation: people write *mn* for

Checkpoint	ratio	bits	token-count normaliser		per-byte loss		norm. %
			bits	factor	bits	factor	
Qwen3.5-9B	58	5.86	-0.34	0.8	6.20	73.6	-6
Qwen3.5-9B-Base	58	5.87	-0.34	0.8	6.21	74.0	-6
Qwen3.5-4B-Base	63	5.98	-0.35	0.8	6.33	80.6	-6
Qwen3.5-4B	64	6.01	-0.36	0.8	6.36	82.4	-6
Gemma-7B●	149	7.21	-0.20	0.9	7.42	171.1	-3
Mistral-7B-v0.3●	175	7.45	2.72	6.6	4.73	26.6	37
Zephyr-7B-beta●	183	7.52	3.08	8.4	4.44	21.7	41
Qwen1.5-1.8B●	196	7.62	2.83	7.1	4.79	27.6	37
Qwen2-7B●	216	7.75	2.68	6.4	5.07	33.6	35
OPT-350M●	226	7.82	4.74	26.6	3.09	8.5	61
TinyLlama-1.1B-Chat●	232	7.86	2.95	7.7	4.90	29.9	38
LaMini-GPT-1.5B●	244	7.93	5.27	38.5	2.67	6.4	66
Qwen2-7B-Instruct	249	7.96	2.74	6.7	5.22	37.3	34
Phi-4-14B●	287	8.16	3.49	11.2	4.68	25.6	43
OPT-1.3B●	290	8.18	4.53	23.1	3.65	12.6	55
OPT-2.7B●	294	8.20	4.42	21.4	3.78	13.7	54
Llama-3.1-8B●	301	8.24	3.07	8.4	5.16	35.8	37
BLOOM-1B1●	323	8.34	4.72	26.3	3.62	12.3	57
Llama-3.1-8B-Instruct	325	8.35	3.28	9.7	5.07	33.5	39
Gemma-2-9B	357	8.48	-0.20	0.9	8.68	411.2	-2
Cerebras-GPT-1.3B●	364	8.51	4.39	21.0	4.11	17.3	52
BLOOM-560M●	389	8.60	4.93	30.6	3.67	12.7	57
Llama-3.2-3B●	417	8.70	3.33	10.1	5.37	41.4	38
Llama-3.2-1B●	419	8.71	3.42	10.7	5.29	39.0	39
StableLM-Zephyr-3B●	466	8.87	5.41	42.6	3.45	10.9	61
BLOOM-3B●	469	8.87	4.35	20.4	4.53	23.0	49
Pythia-410M●	482	8.91	4.45	21.9	4.46	22.0	50
Mistral-Nemo-Base-2407●	691	9.43	4.85	28.8	4.58	24.0	51
Hormoz-8B●	698	9.45	3.72	13.2	5.72	52.9	39
Minitron-8B-Base●	710	9.47	5.56	47.1	3.91	15.1	59
SmolLM3-3B●	769	9.59	4.50	22.6	5.09	34.0	47

Table 6: Equation 3 applied to every checkpoint, sorted by the reported perplexity ratio. “ratio” is Romanized over Sinhala-script perplexity and “bits” its base-2 logarithm, which the two terms partition exactly. The last column is the normaliser’s share of the total. ● marks the 24 checkpoints of the published pool. For the Qwen3.5 and Gemma checkpoints the normaliser term is slightly negative, because their tokenizers spend about the same number of bytes per token in both scripts, and those are also the checkpoints with the smallest reported ratios.

our *man* and *dan* for our *daen*. English loanwords are a smaller but more consequential class, concentrated in the highest-divergence sentences, where a person writes *phone*, *video* or *tv* and our transliterator renders the Sinhala spelling of the same word as *foon*, *wiidiyoo* or *tiwii*.

Table 14 puts a ceiling on how much this matters, using the 450,587-word Swa-bhasha lexicon, which records every Romanized spelling a human supplied for each Sinhala word and averages 15.7 accepted spellings per word. Once the long-vowel convention is normalised, 88.6% of our word tokens in the parallel corpus carry a spelling some human actually supplied for that word, and 87.0% on SOLD. The comparison that matters is the row above: the native speakers who typed the same 500 sentences hit the lexicon 84.2% of the time by the same measure. Our spellings are therefore attested about as often as real typing is.

F.3 What the residual costs a model

Spelling statistics say our output is plausible Singlish. They do not say whether a model finds it easier or harder than the real thing, and that is

what decides whether our downstream gaps are over- or under-estimates. The parallel corpus lets us measure it directly, because it already carries a Romanized side that native speakers typed for the same 500 sentences. We added a third condition, our transliterator’s output for those sentences, and scored all three with the protocol of §4.

Table 16 reports the six checkpoints we were able to rerun. The Sinhala-script and human-typed columns reproduce the recorded run to within 0.11%, which is an independent check on the intrinsic pipeline. The column that matters is the difference: our transliteration costs 0.17 to 0.24 more bits per byte than the human-typed text on identical content, positive for all six. Our Romanized condition is the harder end of the range, so the downstream gaps in §5 are more likely upper bounds than lower bounds.

The same table illustrates the paper’s own thesis in miniature. Per-token perplexity is *lower* on our text than on the human text for all six checkpoints, a median ratio of 0.74, because our spellings are longer and occupy more tokens. Judged by perplexity our transliteration looks easier. Judged per byte

Checkpoint	Sinhala script					Romanized				Mixed script		
	B	PPL	BPB	BPC	BPW	PPL	BPB	BPC	BPW	PPL	BPB	BPW
Llama-3.1-8B	8.0	3.22	1.148	3.044	16.33	970.5	3.584	3.594	22.12	6.36	1.641	19.66
Gemma-2-9B	9.2	13.23	1.190	3.155	16.92	4726.1	3.999	4.010	24.68	48.99	1.792	21.47
Llama-3.2-3B	3.2	3.74	1.296	3.435	18.42	1560.7	3.831	3.842	23.65	7.40	1.776	21.27
Llama-3.1-8B-Instruct‡	8.0	3.86	1.326	3.516	18.86	1255.5	3.718	3.728	22.95	7.94	1.839	22.03
Phi-4-14B‡	14.7	4.01	1.341	3.556	19.07	1149.1	3.446	3.456	21.27	8.08	1.832	21.94
Qwen3.5-9B-Base	9.0	21.39	1.350	3.578	19.19	1250.1	3.290	3.299	20.30	78.56	1.958	23.46
Pythia-410M	0.4	2.58	1.351	3.580	19.20	1245.0	3.707	3.717	22.88	4.91	1.978	23.69
Qwen3.5-9B‡	9.0	22.56	1.373	3.640	19.52	1307.8	3.311	3.320	20.43	84.24	1.990	23.83
Llama-3.2-1B	1.2	4.11	1.387	3.676	19.72	1718.6	3.881	3.892	23.96	8.54	1.903	22.79
Qwen3.5-4B-Base	4.0	23.92	1.399	3.709	19.89	1512.7	3.378	3.387	20.85	83.32	1.985	23.77
Gemma-7B	8.5	21.13	1.406	3.726	19.99	3138.9	3.805	3.816	23.48	66.00	1.930	23.11
Qwen2-7B	7.6	6.29	1.410	3.737	20.05	1357.0	3.498	3.508	21.59	14.11	1.919	22.98
Qwen2-7B-Instruct‡	7.6	6.46	1.431	3.793	20.34	1611.2	3.581	3.591	22.10	15.45	1.985	23.77
Qwen3.5-4B‡	4.0	25.84	1.433	3.799	20.37	1663.1	3.422	3.431	21.12	90.71	2.023	24.23
TinyLlama-1.1B-Chat‡	1.1	4.38	1.469	3.894	20.89	1014.2	3.923	3.934	24.21	8.92	1.980	23.71
Mistral-Nemo-Base-2407	12.2	2.80	1.486	3.938	21.12	1938.4	3.896	3.907	24.04	4.63	1.912	22.90
Cerebras-GPT-1.3B	1.3	2.86	1.491	3.952	21.20	1040.2	3.700	3.710	22.83	5.23	2.052	24.58
Mistral-7B-v0.3	7.2	5.26	1.559	4.133	22.17	921.4	3.884	3.895	23.97	10.15	1.984	23.76
Hormoz-8B‡	8.0	5.98	1.567	4.154	22.28	4175.8	4.045	4.056	24.96	13.58	2.083	24.94
Minitron-8B-Base	8.4	2.98	1.575	4.175	22.39	2118.6	3.428	3.437	21.15	5.08	2.000	23.95
Qwen1.5-1.8B	1.8	8.07	1.601	4.245	22.77	1584.9	3.573	3.583	22.05	17.60	2.079	24.90
BLOOM-3B	3.0	6.95	1.748	4.633	24.85	3258.5	3.626	3.636	22.38	15.90	2.254	26.99
OPT-2.7B	2.7	3.59	1.833	4.860	26.06	1052.8	3.974	3.985	24.53	6.26	2.300	27.54
SmollM3-3B‡	3.1	6.98	1.876	4.973	26.67	5373.0	4.145	4.156	25.58	13.89	2.301	27.56
OPT-1.3B	1.3	3.87	1.942	5.149	27.62	1121.9	4.010	4.021	24.75	6.82	2.407	28.83
Zephyr-7B-beta‡	7.2	8.28	1.986	5.265	28.24	1515.6	4.167	4.179	25.72	15.76	2.360	28.27
BLOOM-1B1	1.1	11.00	2.162	5.732	30.74	3558.5	3.666	3.676	22.62	24.52	2.607	31.22
StableLM-Zephyr-3B‡	2.8	4.57	2.163	5.734	30.76	2129.2	3.986	3.997	24.60	7.51	2.509	30.05
OPT-350M	0.3	4.75	2.238	5.934	31.82	1075.9	3.986	3.997	24.60	8.15	2.630	31.50
BLOOM-560M	0.6	12.64	2.287	6.064	32.52	4915.0	3.810	3.821	23.52	28.01	2.715	32.52
LaMini-GPT-1.5B‡	1.5	5.41	2.395	6.350	34.06	1322.0	3.827	3.838	23.62	9.46	2.788	33.39

Table 7: Corpus-pooled intrinsic metrics for all 31 checkpoints, sorted by Sinhala-script bits per byte. ‡ marks the checkpoints also evaluated downstream.

Model	Sinhala script		Romanized	
	BPB	#LMs <	BPB	#LMs <
character 1-gram	1.815	22	3.977	24
character 2-gram	1.352	7	3.063	0
character 3-gram	1.162	1	2.685	0
character 4-gram	1.269	2	2.636	0
character 5-gram	1.474	15	2.947	0
character 6-gram	1.680	21	3.381	3
best checkpoint	1.148	–	3.290	–
median checkpoint	1.486	–	3.805	–

Table 8: Held-out add-1 character n -gram models against the 31 checkpoints, five-fold cross-validated so every score is out of sample. “#LMs <” counts checkpoints that achieve lower bits per byte than the n -gram model. No checkpoint beats a character bigram on Romanized Sinhala.

Stratum	Items	S %	R %	Δ	p_{Holm}
<i>by difficulty</i>					
Easy	1,851	37.0	29.2	7.9	< .0001
Medium	2,503	34.9	28.7	6.2	< .0001
Hard	2,525	27.7	23.4	4.3	< .0001
<i>by domain</i>					
Social Science	1,059	37.2	26.9	10.3	< .0001
Business Studies	463	34.4	27.1	7.3	< .0001
Other	1,014	31.7	24.6	7.0	< .0001
Stem	614	33.3	26.5	6.7	< .0001
Language	388	29.0	24.4	4.6	< .0001
Humanities	3,341	31.9	27.9	4.1	< .0001

Table 9: SinhalaMMLU by stratum, pooled over the six competent checkpoints. Item counts are per checkpoint. Domains are ordered by loss.

it is harder. The two metrics disagree about the direction of the effect, on the same 500 sentences.

G Licences and Redistribution

- **SinhalaMMLU** (Pramodya et al., 2025): CC BY-NC-ND 4.0, gated. The public release is a 1,851-item sample. We obtained the full 6,879-item set from the authors by email in August 2026, and they asked that it not be re-distributed. The NoDerivatives term also for-

bids distributing adaptations, and our Romanized version is one. We therefore redistribute neither the items nor their romanization. We release the transliteration and set-construction code, the item identifiers, and a SHA-256 digest of the frozen file, so a reader with their own copy of the full set can rebuild ours and confirm it is byte-identical. Everyone else can reproduce the SOLD and Global PIQA results in full.

- **SOLD** (Ranasinghe et al., 2025): publicly released for research through its Hugging Face

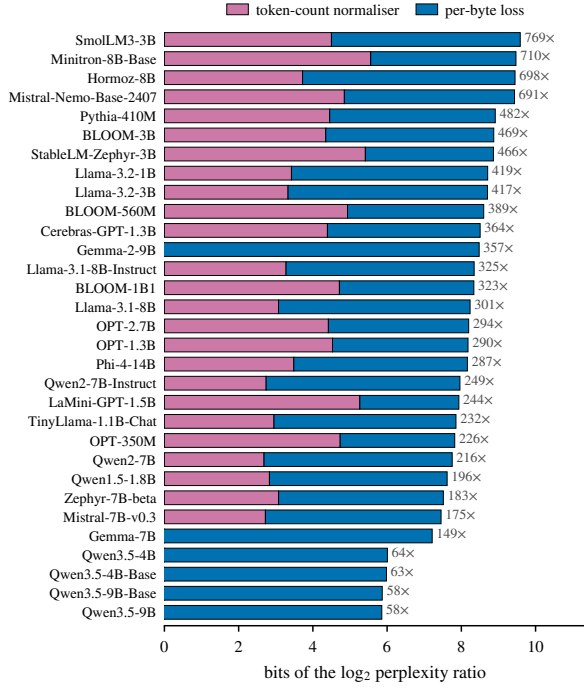


Figure 5: Where the reported degradation ratio comes from, per checkpoint. Each bar is the base-2 logarithm of that checkpoint’s perplexity ratio, split exactly by Equation 3 into the part caused by identical content occupying a different number of tokens and the part caused by a real rise in per-byte loss. The annotation is the raw ratio. The six checkpoints with no visible normaliser term are the Qwen3.5 and Gemma models, whose tokenizers are efficient on the Sinhala script.

Checkpoint	MMLU inv. %		PIQA inv. %		τ
	S	R	S	R	
TinyLlama-1.1B-Chat	1.47	1.03	18	28	0.45
LaMini-GPT-1.5B	95.93	16.35	100	28	0.31
StableLM-Zephyr-3B	3.82	2.63	0	0	0.32
SmolLM3-3B	0.00	0.00	0	0	0.48
Qwen3.5-4B	0.00	0.00	0	0	0.72
Zephyr-7B-beta	2.11	0.64	0	0	0.49
Qwen2-7B-Instruct	0.03	0.01	0	0	0.51
Llama-3.1-8B-Instruct	0.00	0.00	0	0	0.45
Hormoz-8B	0.00	0.00	0	0	0.50
Qwen3.5-9B	0.00	0.00	0	0	0.72
Phi-4-14B	0.00	0.03	0	0	0.42

Table 10: Unparseable-output rates, and τ , the ratio of Romanized to Sinhala-script input tokens on Sinhala-MMLU. τ is below one for every checkpoint, so the Romanized condition is the shorter input in every case.

dataset and its repository, neither of which carries an explicit licence file. We redistribute the 2,500-item test split with our Romanized side attached, in the same pseudonymised form as upstream, and will remove it on request from the authors.

Checkpoint	Acc. %			disc.	
	S	R	Δ	b/c	p_{Holm}
TinyLlama-1.1B-Chat	40	33	+7	12/5	1.0000
LaMini-GPT-1.5B	0	36	-36	0/36	<.0001
StableLM-Zephyr-3B	45	44	+1	15/14	1.0000
SmolLM3-3B	53	51	+2	14/12	1.0000
Qwen3.5-4B	57	49	+8	22/14	1.0000
Zephyr-7B-beta	48	50	-2	18/20	1.0000
Qwen2-7B-Instruct	45	42	+3	21/18	1.0000
Llama-3.1-8B-Instruct	53	48	+5	22/17	1.0000
Hormoz-8B	47	48	-1	13/14	1.0000
Qwen3.5-9B	53	45	+8	25/17	1.0000
Phi-4-14B	46	46	+0	18/18	1.0000

Table 11: Global PIQA, 100 items. “disc. b/c” are the discordant pair counts entering McNemar’s test. No checkpoint is above chance in the Sinhala-script condition and no comparison survives correction, so we treat this task as underpowered rather than as corroboration.

- **Global PIQA** (Chang et al., 2025; de Silva and Ranathunga, 2026): CC BY-SA 4.0, evaluation only. The authors forbid training on it or on synthetic data seeded from it. We do no training. Our Romanized version is a derivative and is redistributed under the same CC BY-SA 4.0 terms.
- **Swa-bhasha Resource Hub** (Sumanathilaka et al., 2025): used only to select and validate the transliterator. We redistribute none of it and our code downloads it from source.
- **The parallel and mixed-script intrinsic corpora** (Rajapakse and Weerasinghe, 2026): released publicly with that paper and redistributed unchanged.
- **Aksharamukha** (Rajan, 2024): AGPL-3.0, version 2.3, called as a library for comparison only. **uroman** (Hermjakob et al., 2018): version 1.3.1.1, MIT-style licence with a required acknowledgement, which appears in the Acknowledgments section as its licence directs.
- **Model weights**: evaluation only, no fine-tuning and no derivative release, which is within the terms of the Llama Community License, the Gemma Terms of Use, the Big-Science RAIL licence and the Apache-2.0 and MIT licences covering the remaining families.

H Reproducibility and Compute

Every number in this paper is produced by scripts that read only the per-item model outputs, so the

Fit	Value
<i>across checkpoints, n = 10</i>	
levels: Romanized on native accuracy	
slope	0.221
95% boot. CI	[+0.113, +0.538]
leave-one-out	0.173 to 0.256
R^2	0.647
p, slope = 0	.0050
p, slope = 1	< .0001
<i>headroom form: loss on headroom</i>	
slope	0.779
95% boot. CI	[+0.457, +0.883]
leave-one-out	0.744 to 0.827
R^2	0.958
p, slope = 0	< .0001
p, slope = 1	.0050
<i>permutation null, Romanized scores shuffled</i>	
levels slope, mean	-0.000
levels R^2 , mean	0.113
levels $\Pr(R^2 \geq \text{observed})$	0.005
headroom slope, mean	+1.000
headroom R^2 , mean	0.936
headroom $\Pr(R^2 \geq \text{observed})$	0.089
<i>across benchmark cells, n = 16</i>	
levels: Romanized on native accuracy	
slope	0.451
95% boot. CI	[+0.359, +0.628]
leave-one-out	0.428 to 0.501
R^2	0.753
p, slope = 0	< .0001
p, slope = 1	< .0001
item-weighted slope	0.476

Table 12: Every refit of the two flattening regressions of §5.2. The levels form regresses Romanized on Sinhala-script accuracy. The headroom form regresses the paired loss on Sinhala-script headroom above chance, and is reported only to show why we do not rely on it: under a permutation null that destroys any association between the two conditions, it still returns a mean slope of 1.00 and a mean R^2 of 0.94, because Sinhala-script accuracy appears on both axes. The levels form returns a mean slope of 0.00 and a mean R^2 of 0.11 under the same null.

tables and figures cannot drift from the data. A checking script re-derives every figure quoted in the text from the frozen analysis and fails if any has moved. We release the frozen evaluation sets with both script conditions paired per record, subject to §G, all per-item generations for every checkpoint, the analysis and figure scripts, and the transliterator: <https://anonymous.4open.science/r/XXXXXXXXX>. The transliterator is also on PyPI, but naming the package would identify us, so the name is withheld until the camera-ready version.

Intrinsic evaluation scores 46,500 sequences, 31 checkpoints at 1,500 sequences each. Downstream evaluation produces 208,538 generations, 11 check-

Method	CER ↓			chrF	Exact	Cov.
	sent.	word	aug.	sent.	word	%
Ours	0.182	0.121	0.112	67.8	33.2	100.00
Web tool $v \rightarrow w$	0.182	0.133	0.114	67.8	30.9	99.67
Aksharamukha	0.191	0.148	0.139	63.7	23.9	100.00
Web tool, raw	0.197	0.176	0.153	63.4	20.6	99.67
uroman	0.228	0.228	0.210	55.7	9.7	100.00

Table 13: Transliteration quality against human references: 4,397 authentic social-media sentence pairs, 450,587 multi-reference words, and a fixed 300,000-sentence sample of a machine-augmented set. Coverage is the share of inputs the method produced any output for.

Text	tokens	cov. %	attested %	
			raw	norm.
parallel, typed by people	2,797	96.1	83.4	84.2
parallel, ours	2,905	96.0	64.3	88.6
SOLD, ours	38,544	90.3	59.5	87.0
Global PIQA, ours	2,244	96.3	63.9	86.9

Table 14: Share of word tokens whose Romanized spelling is one a human supplied for that Sinhala word in the Swa-bhasha lexicon. “tokens” counts the word tokens the lexicon covers and “cov.” the share of all Sinhala word tokens that is. “norm.” collapses runs of the same Latin vowel on both sides. The first row is the reference point: it applies the same measure to text native speakers typed.

Ours	Human	n
<i>mee</i>	<i>meh</i>	33
<i>man</i>	<i>mn</i>	30
<i>daen</i>	<i>dan</i>	28
<i>naae</i>	<i>na</i>	26
<i>mata</i>	<i>matta</i>	19
<i>ee</i>	<i>a</i>	15
<i>naae</i>	<i>nah</i>	12
<i>ban</i>	<i>bn</i>	12
<i>nam</i>	<i>nm</i>	11
<i>gaena</i>	<i>gana</i>	10
<i>thamayi</i>	<i>thamai</i>	10
<i>waedak</i>	<i>wadak</i>	9

Table 15: The twelve most frequent word-level disagreements with the human-typed side of the parallel corpus that survive normalising the long-vowel convention. Most are vowel reduction or abbreviation on the human side.

points over 9,479 items in two script conditions. Both use fp16. Total wall-clock time was about 80 hours: roughly 10 hours for the intrinsic runs and 70 for the downstream runs, of which about 25 hours were Phi-4-14B alone. All runs used commodity dual T4 GPUs except Phi-4-14B, which ran on a single 16 GB RTX 4070 Ti SUPER. The

Checkpoint	bits per byte				PPL	
	B	Sinh.	human	ours	Δ	ratio
OPT-350M	0.35	2.238	3.986	4.195	+0.208	0.85
Pythia-410M	0.41	1.350	3.707	3.937	+0.230	0.80
BLOOM-560M	0.56	2.287	3.810	4.029	+0.220	0.62
TinyLlama-1.1B-Chat‡	1.10	1.468	3.922	4.096	+0.174	0.73
Qwen1.5-1.8B	1.80	1.601	3.573	3.791	+0.218	0.73
SmolLM3-3B‡	3.10	1.878	4.147	4.387	+0.240	0.75
median	—	—	—	—	+0.219	0.74

Table 16: Our transliteration against human typing on the same 500 sentences. Δ is ours minus human bits per byte, positive meaning our text is harder. “PPL ratio” is our perplexity over the human perplexity, below one for every checkpoint even though bits per byte is higher, because our spellings occupy more tokens. The Sinhala-script and human columns reproduce the recorded run to within 0.11%. ‡ marks checkpoints also evaluated downstream. Six of 31 checkpoints, all below 3.2B and none passing the competence screen, so this bounds the direction rather than settling it.

Intrinsic predictor	MMLU gap		SOLD Δ_{MCC}	
	ρ	95% CI	ρ	95% CI
Sinhala BPB	−0.83	[−1.00, −0.30]	−0.78	[−1.00, −0.36]
Romanized BPB	−0.85	[−1.00, −0.43]	−0.88	[−1.00, −0.40]
Sinhala BPW	−0.83	[−1.00, −0.31]	−0.78	[−1.00, −0.36]
BPB gap	−0.15	[−0.89, +0.63]	−0.42	[−0.83, +0.33]
PPL ratio	−0.52	[−0.94, +0.28]	−0.41	[−0.96, +0.40]
parameters	+0.58	[−0.06, +0.91]	+0.47	[−0.14, +0.90]

Table 17: Spearman correlations between intrinsic measurements and downstream script effects, over the ten parseable checkpoints, with bootstrap intervals. Levels track the effect, gaps and ratios do not. The intervals are wide because $n = 10$, which is why this is exploratory.

Attest.	items	att. %	S %	R %	Δ	95% CI
low	822	40.8	61.0	56.0	5.0	[3.4, 6.6]
middle	925	60.7	61.9	54.7	7.2	[5.6, 8.8]
high	700	78.9	60.4	53.2	7.3	[5.4, 9.1]

Table 18: SOLD items split into terciles by how often their Romanized spellings are attested in the human lexicon, pooled over the five competent checkpoints. If our transliterator’s idiosyncrasies drove the Romanized penalty, the low-attestation tercile would lose most. It loses least, and per item the correlation between attestation and loss is Spearman 0.02 ($p = 0.31$, $n = 2,447$). The tercile with the smallest gap also has the lowest Sinhala-script accuracy, which the flattening of §5.2 already predicts.

non-determinism, and the transliteration and data preparation steps are fully deterministic with byte-identical outputs recorded as SHA-256 digests.

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transliteration control of §F.3 adds 9,000 scored sequences and ran in under an hour on eight CPU cores in fp32, which is why its reproduction deviations are non-zero.

Greedy decoding and a fixed template make the downstream runs deterministic up to GPU